Vision–Language Models on the DriveJudgeBench driving-video benchmark: a ten-model audit of failure modes and the agentic gap

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May 2026

Abstract

We extend two prior single-model audits (GPT-5.4-mini, Gemini 3 Flash) into a uniform cross-model failure analysis of **ten** vision–language configurations on the DriveJudgeBench driving-video benchmark (7371 (video, question) pairs, six categories, five JPEG frames at 1 fps per clip). Eight new systems are added: six open-source baselines (Cosmos-Reason, InternVL3.5-8B/30B, LLaVA-OneVision, Qwen3-VL, Qwen3-Omni) and two agentic Qwen3-VL configurations (with and without hint scaffolding). Overall accuracy ranges from 21.9 % (Qwen3-VL) to 64.0 % (Gemini 3 Flash). Open-source baselines cluster between 22 % and 31 %; the bare agent (v4o) is at 60.1 % and the hinted agent (v4p) at 50.9 %. Per-category analysis isolates six failure modes that recur across ≥ 5 models: a near-deterministic "Real" prior on generation detection, Likert mode-collapse on the midpoint or upper bin, a constant-"Yes" classifier on violation-heavy traffic-rules questions, continuous-motion attribution that denies generative artefacts, ego-relative reference-frame inversion on overtake-side, and absence-of-evidence reasoning that infers compliance from the lack of a visible incident. The agent route shifts the regime by replacing language-prior reasoning with a frequency-domain anisotropy threshold (FFT) and an optical-flow priority rule (SOP-C) that promote tool-measured cues above heuristic checklists; hint scaffolding helps only on traffic-laws (where it furnishes the violation vocabulary the bare agent lacks) and otherwise re-imports the absence-of-evidence trap. Gemini 3 Flash matches the agent without tools by combining a learned generative-artefact taxonomy ("morphing", "mushy textures", "physically implausible lens flare") with frame-pair object-permanence reasoning and strong OCR on near-English signage — a profile no open baseline currently produces.

1 Setup

Benchmark. The harness joins each model's per-record output against the two ground-truth question files (`questions_cosmos-drive-dreams` and the `cosmos-predict1` sibling), parses an answer from the benchmark's mandated *"Feedback::: / Evaluation: / Answer: ..."* response shape, and computes accuracy plus per-source, per-category, per-question and per-cell confusion tables. For the six open-source baselines whose `vlm_answer` column was stored as the unparsed full output we re-extract the final "Answer:" token via a last-occurrence regex with question-aware Yes/No, 1–3, Real/Generated and Left/Right fallbacks. The agentic v4o/v4p systems already emit a structured `final_answer` field. The pipeline script writes one enriched parquet per model plus a single accuracy-summary JSON.

Sources. Videos come from two splits. *cosmos-drive-dreams* (6385 answered questions per system, folders `negative/`, `aggressive_takeover/`, `crossing_red_lights/`, `crossing_stop/`) contains photoreal world-model output. *cosmos-predict1* (986, folder `artifacts/`) contains clips with visible diffusion artefacts and supplies the entire Artefacts-understanding category, all 727 of its items. Reality understanding draws from both sources (2186 drive-dreams + 986 predict1). Every

Reality-binary item has GT = *Generated*; the realism Likert is in $\{1, 2\}$; the safety Likert is in $\{1, 2, 3\}$; the artefacts and traffic-rules questions are heavily skewed (88 % GT = Yes on artefacts, 99 % GT = No on "following traffic rules"). These priors are exploitable, and several models do exploit them, which we flag throughout.

Models. Ten configurations, eight evaluated in this work and two contributed by prior single-model audits we synthesise alongside. We refer to them by short identifier (Tables 1, 2).

2 Headline accuracy

Table 1 sorts the ten configurations by overall accuracy. Three regimes coexist: (i) open-source baselines clustered in the 22 %–31 % band, (ii) two closed-source single-pass models (GPT-5.4-mini at 39.1 %, Gemini 3 Flash at 64.0 %), (iii) the agentic Qwen3-VL pipeline (60.1 % bare, 50.9 % hinted) which matches Gemini on the categories where its tools fire but trails it on near-saturated single-frame questions. Table 2 breaks the score down by category and exposes that no model dominates uniformly: even Gemini sits below the agentic systems on "Has the ego car stopped?" (Spatial-temporal) and below GPT-5.4-mini on Visual understanding.

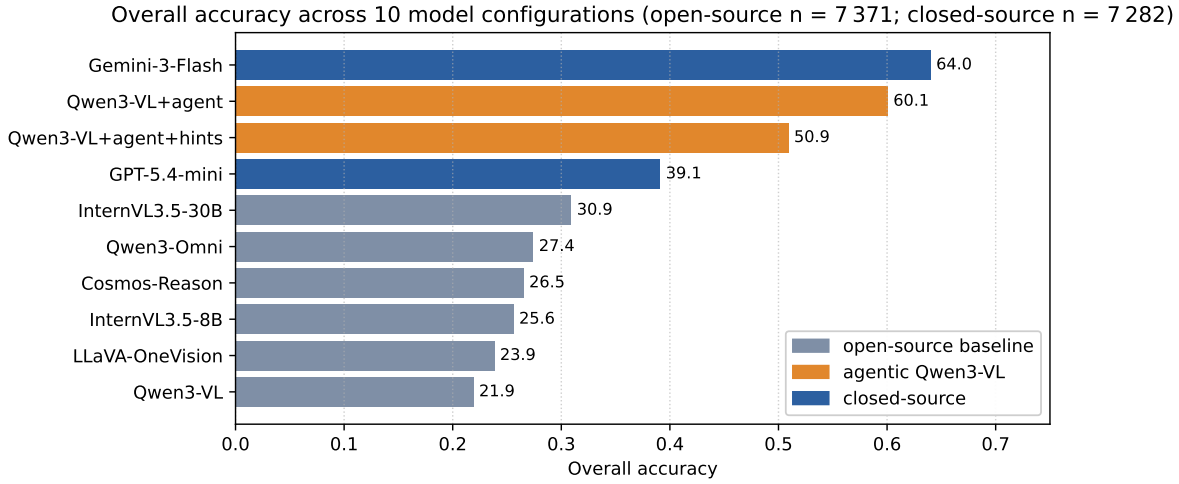


Figure 1: Overall accuracy across the ten configurations on the 7371-pair evaluation set. Open-source baselines (grey) cluster in the 22 %–31 % band; the agentic Qwen3-VL pipeline (orange) sits at 60.1 % bare and 50.9 % hinted; closed-source models (blue) sit at 39.1 % (GPT-5.4-mini) and 64.0 % (Gemini 3 Flash).

3 Cross-model failure modes

Six failure modes recur across ≥ 5 of the ten configurations and account for the bulk of the absolute error count. We list them in decreasing order of total error mass.

(M1) "Real" prior on generation detection. Across the six open-source baselines, the binary *Is it a real video or generated?* produces between 1291 and 1597 false negatives out of 1597 items, i.e. a false-negative rate from 80.8 % (Cosmos-Reason) to 100.0 % (LLaVA-OneVision; 99.3 % for Qwen3-VL). The reasoning is uniformly absence-of-artefact reasoning: *"no obvious signs of digital artifacts or inconsistencies that would suggest it is generated by AI"* (Qwen3-VL, `negative/agg_tk_030_Foggy.mp4`); *"Generated videos often have subtle but noticeable artifacts… this video does not display such signs"* (InternVL3.5-30B, `cross_red_036_Original.mp4`). The "mental checklist" (warping, flicker, duplication) is tuned to GAN-era tells while the clips are diffusion-based world-model output specifically

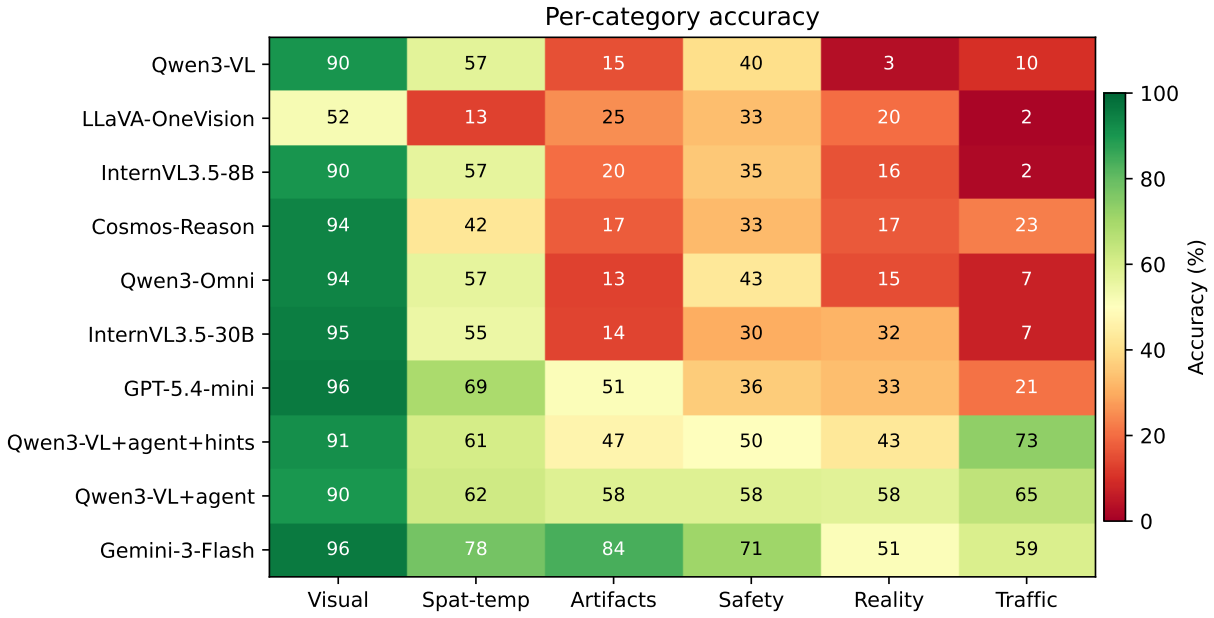


Figure 2: Per-category accuracy. Visual understanding is near-saturated for nine of ten configurations (LLaVA-OneVision is the outlier); Reality / Artefacts / Traffic-laws are where the open-source baselines collapse and where the agent route or Gemini's pre-trained artefact taxonomy provides the lift.

engineered to remove them. GPT-5.4-mini and the bare Qwen3-VL show the same profile (FN rates 88.9 % and 99.3 %); Gemini cuts it to 44.6 %; the agent route cuts it to 25.4 % (v4o) by replacing the language prior with a frequency-domain anisotropy test (§5).

(M2) Likert mode-collapse. The 1–3 Likert questions (realism and safety) collapse to one or two of three labels in nine of ten configurations. Cosmos-Reason, InternVL3.5-30B, InternVL3.5-8B and LLaVA-OneVision each emit "2" on $\geq$81 % of all safety-Likert items (Cosmos: 970/1082; InternVL-30B: 1011/1082); since the safety GT is dominated by the extremes (485 GT = 1, 597 GT = 3), a constant-2 strategy floors them at 8 %–13 %. Qwen3-VL and Qwen3-Omni collapse onto {2,3} but never emit "1". GPT-5.4-mini emits "1" zero times on the safety Likert. Gemini emits "2" essentially never (31 GT = 2 cases out of 1553 makes the middle class nearly untestable, but its predictions concentrate on {1,3}, with 245 of 334 score-task errors being the maximally-wrong GT = 1 $\rightarrow$ VLM = 3). The bare agent (v4o) is the only system that emits "1" at a meaningful rate (121 times) and the only one whose safety-Likert accuracy (50.9 %) exceeds the binary floor; the hinted agent regresses to 16 ones (§6).

(M3) Constant-"Yes" on traffic-rules. On *Is the ego car following the traffic rules?* (n = 512, GT = No in 510), the six baselines answer Yes between 82 % and 100 % of the time on GT = No clips: LLaVA-OneVision 655/655 (recall on No is exactly 0), InternVL3.5-8B 652/665, Qwen3-Omni 618/655, InternVL3.5-30B 617/655, Qwen3-VL 600/655, Cosmos-Reason 503/645. GPT-5.4-mini and Gemini show the same direction at lower magnitude (82.2 % and 52.4 % FP-on-No respectively). The mechanism is a templated checklist ("safe distance / within lane / following traffic flow") reached by absence-of-evidence reasoning: *"no visible signs of reckless driving"* (Qwen3-Omni, `agg_tk_017_Night.mp4`). The violation vocabulary is restricted to overt, single-frame events (speeding, lane change, red-light, stop sign), so trajectory-encoded violations (cut-offs, tailgating, shoulder driving, rolling stops) never trigger a No.

(M4) Continuous-motion attribution on artefacts. On *Does any object appear or disappear without continuous motion?*, the phrase *"continuous motion"* appears in 33 %–41 % of every

Model	n	Accuracy
Qwen3-VL	7371	21.92 %
LLaVA-OneVision	7371	23.85 %
InternVL3.5-8B	7371	25.61 %
Cosmos-Reason	7371	26.54 %
Qwen3-Omni	7371	27.43 %
InternVL3.5-30B	7371	30.93 %
GPT-5.4-mini †	7282	39.12 %
Qwen3-VL + agent + hints (v4p)	7371	50.93 %
Qwen3-VL + agent (v4o)	7371	60.07 %
Gemini 3 Flash †	7282	64.00 %

Table 1: Overall accuracy on the 7 371-pair evaluation set. Daggered rows are reproduced from prior single-model audits (joined-record counts differ because of OpenAI batch-API failures and a small number of unparseable Gemini calls).

Model	Visual (304)	Spatial-temp (339)	Artifacts (727)	Safety (2164)	Reality (3172)	Traffic (665)
Qwen3-VL	90.5	57.2	15.3	39.9	3.5	9.8
LLaVA-OneVision	52.3	13.0	25.2	33.4	20.1	1.5
InternVL3.5-8B	90.5	56.6	20.1	35.4	15.6	2.0
Cosmos-Reason	94.1	42.5	17.5	32.6	17.0	23.0
Qwen3-Omni	94.1	56.6	13.1	42.7	15.1	7.1
InternVL3.5-30B	95.1	54.9	13.6	30.3	31.6	7.2
GPT-5.4-mini	95.7	69.0	50.9	36.3	33.5	20.8
v4p (agent + hints)	91.4	60.8	46.8	50.2	42.8	**73.1**
v4o (agent)	90.1	61.9	**58.5**	**58.3**	**57.5**	65.0
Gemini 3 Flash	**95.7**	**77.6**	**83.9**	**71.0**	51.1	59.4

Table 2: Per-category accuracy (%) for the ten configurations. Item counts in the header. Column-bests in bold; ties broken in favour of the model that does not use tools. Note that on Visual and Spatial-temporal, every model except LLaVA is near-saturated, so the column-best is essentially noise.

base-model failure file (Qwen3-Omni 257/632, LLaVA 231/544, Cosmos-Reason 239/600). Steady forward camera flow is read as a license to deny any inter-frame discontinuity: *"all objects ... are visible throughout the sequence without sudden appearances or disappearances"* (Qwen3-VL, `cross_stop_003_Night.mp4`). The related *stable-silhouette* ⇒ *no artefact* heuristic ("maintain their shape" in ∼ 40 %–45 % of Qwen-family failures) collapses identity-preservation onto rigidity, missing texture flicker, identity drift and material inconsistency. Together these two heuristics produce constant-No classifiers in five of six baselines (Qwen3-VL emits exactly zero "Yes" across 616 artefact failures; InternVL3.5-30B emits 1).

(M5) Reference-frame inversion on overtake-side. On *On which side is the ego car overtaking?* every model that commits to a side flips it. Across the eight in-house systems plus GPT-5.4-mini and Gemini, the modal failure is a coherent ego-relative description in the chain-of-thought followed by an inverted label: *"moving from the right lane to the left lane to overtake them"* → Left, GT Right (InternVL3.5-30B, `agg_tk_015_Night.mp4`); *"passes a black truck on the right between frames 1 and 2 ... overtaking on the left"* (Gemini, `agg_tk_017_Rainy.mp4`). The benchmark labels the lane the ego *enters* (or equivalently the side of the overtaken vehicle); the models label the side of the ego relative to the overtaken vehicle. The two conventions are opposites, so every coherent perception flips. The agent route does not help (v4o 12.0 %; v4p 6.8 %) because the failure is downstream of perception — the labelling step, not the optical-flow tool. A single-sentence prompt

clarification would recover most of this category.

(M6) Absence-of-evidence as compliance / safety. The unifying error pattern across Reality, Safety, Traffic-laws and Artefacts is the inference *absent visible violation* ⇒ *compliant / safe / authentic / artefact-free*. Examples appear above for each category; the structural pattern is that all ten configurations enumerate what is *not* present and conclude the non-violation answer. The 5-frame/1-fps temporal budget makes this worse: every category whose answer requires inter-frame reasoning (stop-line traversal, takeover dynamics, brief artefact pop) interpolates a compliant state-change between sampled frames rather than flagging the ambiguity. This single mode is the largest absolute error source in every system except the bare agent.

4 Per-category briefs

The per-category audits behind these summaries are in `baseline_analysis/reports/*.md`; here we keep only the cross-model contrasts that paper readers need.

Reality understanding (3172 items; 2186 drive-dreams + 986 predict1). Five of six baselines sit between 15 % and 32 %; Qwen3-VL is at 3.5 %. InternVL3.5-30B is the strongest non-agent at 31.6 %, but the lift is entirely Likert mode-collapse on "2": 56.4 % on the realism Likert versus 7.3 % on the binary. Cosmos-Reason has the highest binary accuracy among baselines (19.0 %) thanks to a working — if low-recall — generative-artefact taxonomy that occasionally catches *"the unnatural behavior… could indicate something artificial"* (Cosmos-Reason, `7f3bbe35..._Morning.mp4`). Per-source asymmetry is universal: every system scores worse on *cosmos-predict1* than on *cosmos-drive-dreams*, by 5–30 points (LLaVA: 29.2 % vs 0.1 %; InternVL3.5-30B: 42.5 % vs 7.6 %; v4o agent: 62.1 % vs 47.5 %). Counter-intuitively the lower-fidelity source is harder to call "Generated": the binary GT is the same on both, but cosmos-predict1 clips lack the world-model's frequency-domain signature that triggers the agent's FFT cue, and their visible morphs are exactly what every baseline's stable-silhouette excuse dismisses. The agent's binary accuracy is 74.6 % and is driven almost entirely by an FFT anisotropy threshold (§5); Gemini's 55.4 % is driven by OCR on near-English signage (*"the morphing of the 'STOPP' text on the road"*, *"the 'PARKING' sign … morphing into 'PANNI' "*, *"PALCONERI"*, *"STNULL - STNAID"*).

Safety understanding (2164 items). Six baselines cluster between 30.3 % and 42.7 %; the binary task is essentially a "Yes" classifier for all of them (50.6 %–57.9 %, only marginally above the 52.6 % Yes-prior). The Likert is the divider: v4o jumps to 50.9 %, every baseline stays at 8 %–29 %. The mechanism behind v4o's lift is concentrated: `crossing_red_lights/` Likert goes from 0.7 % to 58.6 % and `negative/` from 47.6 % to 78 %, while `crossing_stop/` (8.4 % → 8.4 %) and `aggressive_takeover/` (0.0 % → 0.6 %) are unchanged. Gemini's 71.0 % matches the agent on red-lights and stops via static-cue reading and beats it on `negative/` clips, but collapses on `aggressive_takeover/` to 19.3 % for the same trajectory-blindness reason as the baselines.

Traffic laws (665 items). Open baselines collapse to 1.5 %–23.0 %. Cosmos-Reason is the strongest (23.0 %) because it is the only baseline that emits "No" more than 100 times ($n = 145$ vs. 3–40 for peers) and has a working stop-sign rule chain. The agent route lifts Qwen3-VL by 54 points (9.8 % → 65.0 %) entirely on red-lights and stops; the hinted agent (v4p) climbs further to 73.1 % — **the only category in the benchmark where v4p beats v4o** — because the violation-targeted hint payload furnishes the missing "cut-off / unsafe overtake" vocabulary (§6). `aggressive_takeover/` accuracy on follow-rules: 5.6 % (baseline) → 5.6 % (v4o) → 85.6 % (v4p), at the cost of −22.5 % on red-lights and −20.9 % on stops.

Artifacts understanding (727 items, GT = Yes in 88%). Five of six baselines sit in the constant-No band (13 %–20 %); LLaVA's 25.2 % is mechanically explained by a 17 % Yes-rate (precision-on-Yes 90.2 % matches the GT prior), not detection. InternVL3.5-30B underperforms its 8B sibling (13.6 % vs 20.1 %) because the larger model's RLHF alignment sharpens the "don't fabricate" prior — the wrong default on a Yes-skewed benchmark. The agent reaches 58.5 % via FFT anisotropy and SAM-mask rigidity-violation vocabulary (*"texture swimming"*, *"components morphing"*, *"wheels warp and merge into the body"*). Gemini reaches 83.9 % *without tools* via per-frame OCR on garbled signage, frame-pair object-permanence reasoning ("between frames 4 and 5 . . .") and a learned named taxonomy (*"mushy textures"*, *"physically implausible lens flare"*). It is the single category where Gemini beats the agent by a wide margin (§7).

Spatial-temporal (339 items). Two questions: "Has the ego car stopped?" (n = 201; six of eight near-saturated, LLaVA at 8.5 %, Cosmos-Reason at 52.7 %) and "On which side is the ego car overtaking?" (n = 133, GT = Right in 127). The overtake-side question is mode M5 in pure form. LLaVA's stop-question collapse (8.5 %) is a constant-Yes anchored to the visible STOP sign rather than to inter-frame motion: *"the ego car has stopped . . . as indicated by the 'STOP' sign visible in the video"* (`cross_stop_055_Night.mp4`); the static landmark is read as evidence of stopping. Cosmos-Reason is the strongest non-agent on overtake-side (25.6 %) precisely because it is the only model that commits to a side rather than hedging — 86 confident Right→Left flips deliver 29 chance hits on the right-skewed GT, versus 102/123 hedges for InternVL3.5-8B that score zero.

Visual understanding (304 items). Near-saturated for seven of eight in-house systems (90 %–95 %). LLaVA-OneVision is the outlier at 52.3 %, driven by a complete collapse on *"Are the traffic lights red?"* (12.5 %, 18/144) where the model emits a near-template *"the traffic lights are not red; they are green"* response on red-light clips, occasionally hallucinating *blue* lights (`cross_red_009_Snowy`). The behaviour is consistent with a model that fails to ground the colour token on the bulb pixels and falls back on a "moving car ⇒ green" prior. The agent route does not help (v4o 90.1 %, below every non-LLaVA baseline) and introduces format-level failures: 18/30 of v4o's visual misses are *NO MATCH!* traces where the model loops the optical-flow tool on trivial colour-cue questions and never closes `final_answer`.

5 Why does the agent route work?

The bare Qwen3-VL is the worst model on the benchmark (21.9 %); the agentic configuration that wraps it is the second-best (60.1 %). The lift is concentrated in exactly the categories the base model is blind to (Reality 3.5 % → 57.5 %, Artefacts 15.3 % → 58.5 %, Traffic-laws 9.8 % → 65.0 %, Safety-Likert 26.7 % → 50.9 %). Three tool-driven mechanisms account for the gain.

FFT anisotropy threshold. A frequency-domain energy-ratio test with a fixed cut-off (0.048) is the decisive cue on Reality. Across the 1597 binary items, 1189 produce anisotropy > 0.048 and v4o answers *Generated* on 1180 of those (99.2 % routing fidelity). Successful calls quote the score directly: *"FFT anisotropy 0.0519 exceeds the 0.048 threshold . . . directional energy patterns typical of synthetic videos"* (`negative/f059738a..._Snowy.mp4`). The same tool also fires on Artefacts: 382 of 425 correct shape-change calls cite the FFT threshold. Note that the test does not measure artefacts directly — it provides a per-clip prior on AI-genness which the VLM combines with SAM-mask object-level inspection.

SAM rigidity-violation vocabulary. SAM segmentation forces the VLM to reason over object crops in isolation, breaking the global "scene looks coherent" frame that drives every base model's stable-silhouette excuse (M4). The reasoning vocabulary that appears in v4o success cases is consistently of the form *"the roof of the car deforms and melts, the body wobbles unnaturally, the wheels warp and merge into the body"* (`jKrJezTsVUk_65520_14b.mp4`). 378 of 425 correct artefact calls

cite at least one of *texture swimming / component morphing / wobbling / melting / merging*. Base Qwen3-VL emits zero such phrases across its 616 artefact failures.

Optical-flow priority rule (SOP-C). On Safety and Traffic-laws, the agent's prompt encodes a priority rule: when the optical-flow tool returns forward-motion (pink/magenta pattern) and a static violation cue (red light, stop sign, stop line) is in frame, the verdict must be *No / unsafe / violating*. This single rule is what breaks the absence-of-evidence trap (M6). On 103 `crossing_red_lights` clips v4o newly catches at Likert 1, the base Qwen3-VL emitted "2" 61 times and "3" 42 times — never "1". The rule does not measure danger; it provides a single binary "ego is moving" cue that, combined with red-light visibility from the original frames, breaks the "no abrupt maneuver = safe" heuristic.

Where the agent does *not* help. (i) Visual understanding: 90.1 % for v4o vs. 95.1 % for the strongest baseline, with 18/30 v4o misses being `NO MATCH!` format failures where the agent loops the optical-flow tool on trivially single-frame questions. (ii) Spatial-temporal overtake-side: 12.0 % (worse than every baseline that commits) because the labelling-convention failure (M5) is downstream of the tool call. (iii) `aggressive_takeover/` on Traffic-laws: 5.6 % unchanged from baseline, because no measured cue in the agent's tool stack captures unsafe-overtake or cut-in dynamics; only the violation-targeted hint payload (v4p) recovers this folder.

6 Hint scaffolding helps only on traffic laws

The hinted agent (v4p) furnishes the VLM with subsidiary questions prior to the verdict (the payload lives in `hint_questions.json` under `data_preparation/resources/`). v4p drops overall accuracy by 9.1 points relative to v4o (60.1 % → 50.9 %) but improves Traffic-laws by 8 points (65.0 % → 73.1 %). The mechanism is consistent across categories: hints help wherever the failure is a *missing-vocabulary* problem and hurt wherever the failure is a *language-prior-overruling-tool-output* problem.

Hints help on traffic laws. The hint payload for `aggressive_takeover` asks *"Did the ego car cross a solid white or double yellow line to pass? Did the ego car cut off the other vehicle when merging back?"*; for `crossing_red_lights` *"Did the ego car's front bumper cross the stop line while the light was red?"*; for `crossing_stop` *"Did the ego car come to a complete halt at the sign?"*. These prompts furnish the violation vocabulary that all baselines and even v4o lack. Result on `aggressive_takeover`: 5.6 % → 85.6 % (+128 correct), at the cost of −22.5 % on red-lights and −20.9 % on stops where the model over-applies the "stop-sign yielding" template; net +8 points.

Hints hurt on reality, artefacts, safety. On Reality, the hint prompt invites the model to weigh visual evidence against the FFT score: on binary items where FFT > 0.048, v4o answers *Generated* on 99.2 %; v4p drops to 84.6 %. The override is verbatim: *"The FFT analysis indicates a high anisotropy score (0.0496), suggesting potential AI generation. However, the SAM segmentation masks show no signs of rigidity violations ... the visual evidence of physical consistency outweighs the statistical anomaly"* (v4p, `negative/cross_stop_014_Original.mp4`). On Artefacts, the appear/disappear hint set is exactly two strict cues (*"Do vehicles 'pop' into existence in the middle of the road? Do distant cars flicker in and out of visibility?"*); the model uses them as a closed checklist and answers No whenever no literal pop or flicker is present, regardless of FFT/SAM evidence. On Safety, the hint sub-questions absorb the agent's attention budget and the optical-flow priority rule is no longer the conclusion the model commits to: v4o emits "1" 121 times, v4p only 16; v4p's red-light Likert collapses 58.6 % → 5.9 %.

Diagnostic. Hints are a textbook example of cue dilution when the bare agent's tools already provide a working signal. They are a useful prompt-engineering lever only when the failure is the

absence of vocabulary (Traffic-laws). The same conclusion follows in the inverse direction: every category where v4o already beats baselines via a tool-measured cue is a category where v4p re-imports the heuristic the tool was designed to suppress.

7 Why does Gemini 3 Flash outperform open baselines and match the agent?

Gemini reaches 64.0 % overall, 83.9 % on Artefacts, 77.6 % on Spatial-temporal, 71.0 % on Safety, and only 51.1 % on Reality (its weakest category). It does so without an FFT/SAM/RAFT toolchain. Five mechanisms recur across its chains-of-thought.

Generative-artefact taxonomy. Gemini's reasoning consistently invokes a small, named vocabulary — *morphing, temporal flickering, physically implausible lens flare, mushy textures, structural integrity, sliding pedestrians* — that maps directly onto the failure modes cosmos-drive-dreams and cosmos-predict1 actually produce. This looks like a learned prior from generative-model evaluation data rather than generic image description. Open baselines *describe* scenes; Gemini *diagnoses* them. Cosmos-Reason is the only open baseline whose reasoning occasionally reaches a similar register (e.g. *"cars seemingly passing through each other"*), which explains its modest lead on the Reality binary.

OCR + language prior on near-English text. The single most cited cue across Gemini's correct Reality / Artefacts calls is near-English text on signs, license plates and pavement markings: *"STOPP"* on the road, *"PARKING"* → *"PANNI"*, *"STNULL - STNAID"*, *"PALCONERI"*, *"ITO"*. This requires both legible character extraction from low-res frames *and* a language model that flags the strings as not-real-words. Open 7B–8B baselines fail on either side of that pipeline. It is by far Gemini's sharpest tool in the artefact-detection toolbox and the main reason it beats the agent on Artefacts (83.9 % vs 58.5 %): the agent's FFT/SAM stack is strong on global frequency anomalies and per-object morph but does not read text.

Frame-pair object-permanence reasoning. In ∼35 % of correct artefact calls Gemini explicitly localises to a frame index (*"between frames 4 and 5"*), treating the input as five labelled images and asking "does object X have a plausible counterpart in frame $i + 1$?". Example: *"the silver sedan directly in front of the black SUV in the first frame disappears instantly in the second frame"* (`cross_stop_010_Snowy.mp4`). This is object-permanence reasoning, not motion estimation; it generalises remarkably well to the "without continuous motion" question even though the model never sees continuous motion. No open baseline cites a frame index in its reasoning at any meaningful rate.

Static-cue dominance on safety / law. Lane lines, stop signs, red lights, weather, and "is the road clear?" are answerable from a single well-chosen frame. With 5 frames Gemini almost always finds at least one that resolves the question, and combines it with a strong language prior for what makes driving unsafe: *"Driving through a red light, especially in rainy conditions with reduced traction and visibility, constitutes a major safety violation"*. This is also where Gemini overlaps with — and on per-question latency, beats — the agentic pipeline, since the agent pays tool-call cost on trivially-resolvable single-frame questions.

Where Gemini still fails. Gemini's residual error is concentrated on motion-encoded subsets that 1-fps sampling destroys: `aggressive_takeover/` traffic-law accuracy is 9 %, `cosmos-predict1` binary Reality is 36.3 % (vs 65.8 % on `cosmos-drive-dreams`), and the safety-Likert collapses to $\{1, 3\}$ on the same takeover subset (5.7 %). These are exactly the categories an FFT/RAFT tool stack is designed to fix — and the agent does fix them (`crossing_red_lights` Safety-Likert 0.7 % → 58.6 %,

`aggressive_takeover` Traffic-laws 5.6 % → 85.6 % with hints). Bolting the agent's tools onto a Gemini base would target exactly the residual error distribution Gemini exhibits.

Implication. The 4-point gap between bare Gemini and v4o is small, but the per-category breakdown is substantively different: Gemini wins on Artefacts (+25.4), Spatial-temporal (+15.7) and Safety (+12.7); the agent wins on Reality (+6.4) and Traffic-laws (+5.6). The natural follow-up is to combine the two profiles — Gemini's per-frame artefact spotter plus the agent's frequency-domain prior and optical-flow priority rule — on the subsets where each currently wins.

8 Synthesis

Stitching the ten audits together, three observations land hardest on the benchmark design.

(1) Open-source 7B–30B VLMs do not detect modern diffusion artefacts. The combined Reality + Artefacts mass is roughly half of the benchmark's question count and the open-source baselines achieve 3 %–32 % on Reality and 13 %–25 % on Artefacts. The ceiling is set by absence-of-artefact reasoning tuned to GAN-era tells. Two interventions are visible in the data: a frequency-domain prior (FFT anisotropy + 0.048 cut-off) lifts Reality from 3.5 % to 74.6 % on the same VLM, and a learned generative-artefact taxonomy in the model's pre-training (Gemini) lifts Artefacts to 83.9 % without any tool. Both work; the second composes better with downstream reasoning, the first is cheaper to bolt onto an open base.

(2) The benchmark conflates perception with labelling on overtake-side. Every model that commits to a side flips the label. This is not a vision deficit — it is a convention mismatch between "side the ego enters" (benchmark) and "side the ego is on relative to the overtaken car" (model). A one-sentence prompt patch would recover most of the 100+ Right→Left flips for every system that currently commits, which by itself would lift the agent above Gemini on Spatial-temporal.

(3) Tool augmentation pays only where the base VLM is blind. Adding the FFT/SAM/RAFT stack to Qwen3-VL flips it from worst to second-best (+38.2 %), but is worse than the base model on the near-saturated Visual category (90.1 % vs 95.1 % top baseline) and equal-to-worse than baselines on the convention-driven overtake-side question. Hint scaffolding compounds this: it helps only when the missing piece is vocabulary (Traffic-laws `aggressive_takeover` 5.6 % → 85.6 %) and otherwise re-imports the language prior the tool was meant to override (Reality 57.5 % → 42.8 %).

Open question. Gemini's *generative-artefact taxonomy in pre-training* appears to do, with zero tools, what the agent does with FFT, SAM and RAFT — on a strictly larger fraction of the benchmark than the agent helps on. We do not have a closed-source open-baseline pair that isolates this mechanism, but the per-category breakdown (Gemini ≫ v4o on Artefacts, < v4o on Reality) suggests the FFT prior and the text-recognition prior are complementary rather than redundant. A follow-up that runs the agent harness on Gemini — or distils Gemini's named-artefact vocabulary into an open VLM's training mix — is the highest-leverage next step the data points to.

A Reproduction

All numbers above are produced by a single script, `baseline_analysis/run_analysis.py`, which reads each model's parquet under `to_be_copied/results/` (excluding the GPT-5.4-mini output already analysed under `closed_source_evaluation/gpt/`) and writes:

- `baseline_analysis/<model>_enriched.parquet` — one row per (model, video, question) with the re-extracted `vlm_answer`, `ground_truth`, `category`, `source` and `correct` columns.
- `baseline_analysis/all_models_accuracy_summary.json` — overall, per-source, per-category, per-question accuracy and full (ground_truth, vlm_answer) confusion tables for each of the eight in-house models.
- `failures_by_model_category/<model>__<cat>.json` — failure dumps consumed by the per-category reviewers, one file per (model, category) pair.
- `baseline_analysis/reports/<cat>.md` — the six per-category cross-model audits behind §4.

The two prior single-model audits we synthesise are reproducible from the analogous wrappers under `closed_source_evaluation/gpt/` and `closed_source_evaluation/gemini/`.